Embedded Systems

CORTEX-M4 INSTRUCTIONS

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SHIFT OPERATIONS

Register shift operations move the bits in a register left or right by a specified number of bits, the shift length. Register shift can be performed directly by the instructions ASR, LSR, LSL, and ROR and the result is written to a destination register.

01 ASR Instruction

Arithmetic shift right by n bits moves the left-hand 32-n bits of the register Rm, to the right by n places, into the right-hand 32-n bits of the result, and it copies the original bit[31] of the register into the left-hand n bits of the result.
You can use the ASR operation to divide the signed value in the register Rm by 2^n, with the result being rounded towards negative-infinity.
When the instruction is ASRS the carry flag is updated to the last bit shifted out, bit[n-1], of the register Rm.

ASR &ASRS

SIMPLE FIGURE OF ASRS

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Thank's For Listening

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